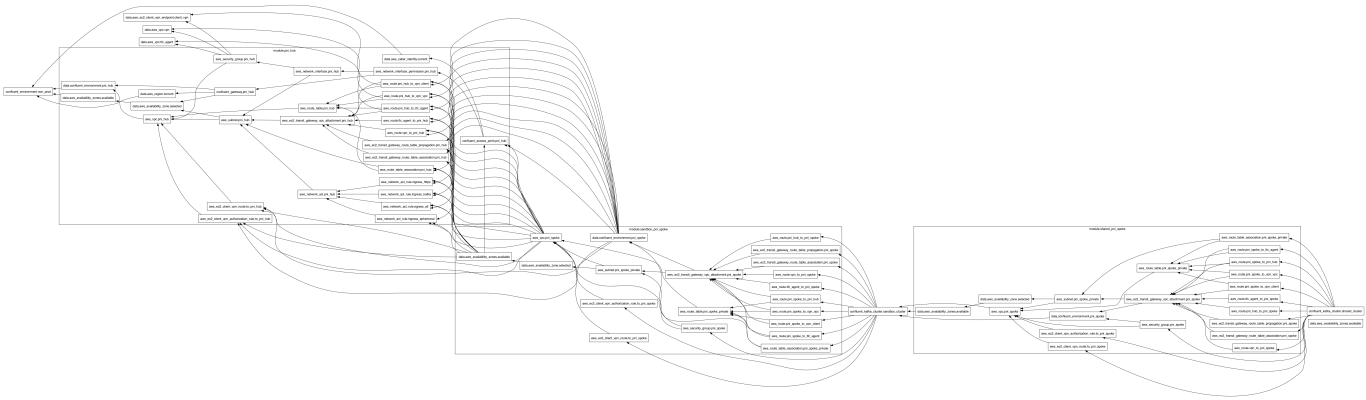
# IaC Confluent Cloud AWS Private Network Interface (PNI), Infrastructure and Networking Example



> **Terraform-managed, hub-and-spoke Private Network Interface (PNI) connectivity between AWS VPCs and Confluent Cloud Enterprise Kafka clusters, deployed via Terraform Cloud.**

---

**Table of Contents**


- 1.0 Overview
- 2.0 Architecture
- 3.0 Module Structure
- 4.0 Prerequisites
  - 4.1 Client VPN, Centralized DNS Server, and Transit Gateway
    - 4.1.1 Key Features Required for Confluent PNI to Work
      - 4.1.1.1 Hub-and-Spoke Network Architecture via Transit Gateway
      - 4.1.1.2 Client VPN Integration
      - 4.1.1.3 Cross-VPC Routing
      - 4.1.1.4 Security & Observability
  - 4.2 Terraform Cloud Agent
    - 4.2.1 Key Features Required for Confluent PNI to Work (TFC Agent Configuration)
      - 4.2.1.1 Custom DHCP Options for DNS Resolution
      - 4.2.1.2 Transit Gateway Connectivity
      - 4.2.1.3 Security Group Configuration for Kafka Traffic
      - 4.2.1.4 AWS VPC Endpoints for Private Service Access
      - 4.2.1.5 ECS Fargate Deployment Pattern
      - 4.2.1.6 IAM Permissions for Infrastructure Management
      - 4.2.1.7 Network Architecture Summary
- 5.0 Configuration
  - 5.1 Key Input Variables
  - 5.2 CIDR Allocations
- 6.0 Deployment
  - 6.1 Create
  - 6.2 Destroy
- 7.0 Outputs

- 8.0 Security Design
- 9.0 How PNI Differs from PrivateLink
- 10.0 Resources


---

## 1.0 Overview

This repo provides a production-grade private networking topology that connects AWS workload VPCs to Confluent Cloud Enterprise Kafka clusters via **Private Network Interface (PNI)**, Confluent's next-generation model with customer-managed ENIs deployed directly in your VPCs. The architecture employs a **Hub-and-Spoke** pattern: a **PNI Hub VPC** owns the ENIs registered with Confluent Cloud and functions as the centralized private connectivity hub; **PNI Spoke VPCs** (sandbox, shared) each run a Confluent Cloud Enterprise Kafka cluster and connect to the hub through AWS Transit Gateway. All VPCs are linked via an existing **AWS Transit Gateway (TGW)**, enabling a VPN-connected developer or operator to access Confluent Cloud endpoints entirely within AWS's private network: *no public endpoints*, *no PrivateLink middleman*, and *no internet gateway required at any point*.

---

## 2.0 Architecture

```
flowchart TD
    subgraph VPN["VPN VPC (existing)"]
        CVE["AWS Client VPN Endpoint"]
    end

    subgraph TFC_AGENT["TFC Agent VPC (existing)"]
        AGENT["Terraform Cloud Agent"]
    end

    TGW["AWS Transit Gateway + Route Table"]

    subgraph PNI_HUB["PNI Hub VPC — 10.3.0.0/20"]
        direction TB
        SG["Security Group (ports 443, 9092 ingress only)"]
        NACL["Network ACL (443, 9092, ephemeral)"]
        ENI1["ENI — AZ-a"]
        ENI2["ENI — AZ-b"]
        ENI3["ENI — AZ-c"]
        TGW_ATT_HUB["TGW Attachment"]
        GW["confluent_gateway (AWS PNI Gateway)"]
        AP["confluent_access_point (PNI Access Point)"]
        SG --> ENI1 & ENI2 & ENI3
        ENI1 & ENI2 & ENI3 --> AP
        AP --> GW
    end

    subgraph SANDBOX_SPOKE["Sandbox PNI Spoke VPC — 10.0.0.0/20"]
        TGW_ATT_SB["TGW Attachment"]
        SK_CLUSTER["confluent_kafka_cluster sandbox_cluster (Enterprise,
HIGH availability)"]
    end
```

```
    subgraph SHARED_SPOKE["Shared PNI Spoke VPC — 10.1.0.0/20"]
        TGW_ATT_SH["TGW Attachment"]
        SH_CLUSTER["confluent_kafka_cluster shared_cluster (Enterprise,
HIGH availability)"]
    end

    subgraph CONFLUENT_CLOUD["Confluent Cloud"]
        ENV["Environment: non-prod (Stream Governance: ESSENTIALS)"]
        GW --> ENV
        SK_CLUSTER --> ENV
        SH_CLUSTER --> ENV
    end

    CVE -->|"routes via TGW"| TGW
    AGENT -->|"routes via TGW"| TGW
    TGW <-->|"attach + propagate"| TGW_ATT_HUB
    TGW <-->|"attach + propagate"| TGW_ATT_SB
    TGW <-->|"attach + propagate"| TGW_ATT_SH
    TGW_ATT_HUB --> PNI_HUB
    TGW_ATT_SB --> SANDBOX_SPOKE
    TGW_ATT_SH --> SHARED_SPOKE
```

---

## 3.0 Module Structure

```
.
├── main.tf                          # Root: environment, cluster
resources, module calls
├── data.tf                          # Data sources: VPCs, VPN endpoint,
AZs
├── variables.tf                     # Root-level input variables
├── outputs.tf                       # PNI gateway & access point IDs
├── versions.tf                      # Provider version pins (AWS 6.33,
Confluent 2.62, TFE 0.73)
├── provider.tf                      # Provider configuration
├── deploy.sh                        # Bootstrap / teardown script (AWS
SSO + TF env vars)
└── modules/
    ├── aws-vpc-confluent-pni-hub/   # PNI Hub module
    │   ├── setup-confluent-pni-hub.tf            # Gateway, access point,
VPC, subnets, ENIs, SG
    │   ├── setup-aws-vpc-tgw-private_routing.tf  # TGW attachment,
associations, routes
    │   ├── setup-aws-vpc-security_group_rules.tf # NACL rules
    │   ├── setup-aws-network-permissions.tf      # ENI permissions for
Confluent's AWS account
    │   ├── data.tf                               # Module-level data
sources
    │   ├── variables.tf                          # Module inputs
    │   ├── outputs.tf                            # Gateway/access point
```

```
IDs exposed to root
    │   └── versions.tf                          # Module provider
constraints
    └── aws-vpc-confluent-pni-spoke/             # PNI Spoke module
(sandbox, shared)
        ├── main.tf
        ├── data.tf
        ├── variables.tf
        ├── outputs.tf
        └── versions.tf
```

---

## 4.0 Prerequisites

This project assumes you have the following prerequisites in place:

- Client VPN, Centralized DNS Server, and Transit Gateway
- Terraform Cloud Agent

### 4.1 Client VPN, Centralized DNS Server, and Transit Gateway

```
%%{init: {'theme': 'base', 'themeVariables': { 'primaryColor': '#1a73e8',
'primaryTextColor': '#fff', 'primaryBorderColor': '#1557b0', 'lineColor':
'#5f6368', 'secondaryColor': '#34a853', 'tertiaryColor': '#fbbc04'}}}%%

flowchart TB
    subgraph USERS["👤 Remote Users"]
        VPNClient["VPN Client
(OpenVPN/AWS Client)"]
    end

    subgraph AWS["☁ AWS Cloud"]
        subgraph VPN_VPC["Client VPN VPC
var.vpn_vpc_cidr"]
            VPNEndpoint["AWS Client VPN
Endpoint"]
            VPNSubnets["VPN Subnets
(Multi-AZ)"]
            VPNSG["Security Group
client-vpn-sg"]
            VPNResolver["Route53 Outbound
Resolver Endpoint"]
            VPNEndpoint --> VPNSubnets
            VPNSubnets --> VPNSG
            VPNSubnets --> VPNResolver
        end

        subgraph TGW["Transit Gateway
signalroom-tgw"]
            TGWCore["TGW Core
ASN: 64512"]
```

```
            TGWRouteTable["Custom Route
Tables"]
            TGWCore --> TGWRouteTable
        end

        subgraph DNS_VPC["DNS VPC (Centralized)
var.dns_vpc_cidr"]
            R53Inbound["Route53 Inbound
Resolver Endpoint"]
            R53PHZ["Private Hosted Zones
*.aws.confluent.cloud"]
            R53Inbound --> R53PHZ
        end

        subgraph TFC_VPC["TFC Agent VPC
var.tfc_agent_vpc_cidr"]
            TFCAgent["Terraform Cloud
Agent"]
        end

        subgraph WORKLOAD_VPCs["Workload VPCs"]
            subgraph WL1["Workload VPC 1"]
                VPCE1["VPC Endpoint
(PrivateLink)"]
            end
            subgraph WL2["Workload VPC N..."]
                VPCEN["VPC Endpoint
(PrivateLink)"]
            end
        end

        ACM["ACM Certificates
(Server & Client)"]
        CWLogs["CloudWatch Logs
VPN & Flow Logs"]
    end

    subgraph CONFLUENT["☁ Confluent Cloud"]
        PrivateLinkService["PrivateLink Service
Endpoint"]
        Kafka["Kafka Cluster
(Private)"]
        PrivateLinkService --> Kafka
    end

    %% Connections
    VPNClient -->|"Mutual TLS
Authentication"| VPNEndpoint
    ACM -.->|"Certificate Auth"| VPNEndpoint

    VPN_VPC -->|"TGW Attachment"| TGW
    DNS_VPC -->|"TGW Attachment"| TGW
    TFC_VPC -->|"TGW Attachment"| TGW
    WL1 -->|"TGW Attachment"| TGW
```

```
    WL2 -->|"TGW Attachment"| TGW

    VPNResolver -->|"DNS Forwarding
Rule"| R53Inbound
    R53PHZ -->|"Returns Private
Endpoint IPs"| VPCE1

    VPCE1 -->|"AWS PrivateLink"| PrivateLinkService
    VPCEN -->|"AWS PrivateLink"| PrivateLinkService

    VPNEndpoint -.->|"Logs"| CWLogs
    TGW -.->|"Flow Logs"| CWLogs

    %% Styling
    classDef userStyle fill:#4285f4,stroke:#1557b0,stroke-
width:2px,color:#fff
    classDef vpcStyle fill:#e8f0fe,stroke:#1a73e8,stroke-width:2px
    classDef tgwStyle fill:#fef7e0,stroke:#f9ab00,stroke-width:3px
    classDef dnsStyle fill:#e6f4ea,stroke:#34a853,stroke-width:2px
    classDef confluentStyle fill:#f3e8fd,stroke:#9334e6,stroke-width:2px
    classDef serviceStyle fill:#fff,stroke:#5f6368,stroke-width:1px

    class USERS userStyle
    class VPN_VPC,TFC_VPC,WORKLOAD_VPCs,WL1,WL2 vpcStyle
    class TGW tgwStyle
    class DNS_VPC dnsStyle
    class CONFLUENT confluentStyle
```

### 4.1.1 Key Features Required for Confluent PNI to Work

#### 4.1.1.1 Hub-and-Spoke Network Architecture via Transit Gateway

- Transit Gateway serves as the central routing hub connecting all VPCs
- Disabled default route table association/propagation for explicit routing control
- DNS support enabled on the TGW (`dns_support = "enable"`)
- Custom route tables for fine-grained traffic control between VPCs

#### 4.1.1.2 Client VPN Integration

- Mutual TLS authentication using ACM certificates (server + client)
- Split tunnel configuration for routing only Confluent traffic through VPN
- Authorization rules controlling which CIDRs VPN clients can access
- Routes added to VPN endpoint for all workload VPC CIDRs via Transit Gateway

#### 4.1.1.3 Cross-VPC Routing

- TGW attachments for: VPN VPC, DNS VPC, TFC Agent VPC, and all Workload VPCs
- Route tables in each VPC with routes to other VPCs via TGW
- Workload VPC CIDRs aggregated and distributed to VPN client routes

#### 4.1.1.4 Security & Observability

- Dedicated security groups per component (VPN endpoint, etc.)
- VPC Flow Logs and TGW Flow Logs to CloudWatch
- VPN connection logging for audit trails
- IAM roles with least-privilege for flow log delivery

## 4.2 Terraform Cloud Agent

```
%%{init: {'theme': 'base', 'themeVariables': { 'primaryColor': '#1a73e8',
'primaryTextColor': '#fff', 'primaryBorderColor': '#1557b0', 'lineColor':
'#5f6368', 'secondaryColor': '#34a853', 'tertiaryColor': '#fbbc04'}}}%%

flowchart TB
    subgraph TERRAFORM_CLOUD["☁ Terraform Cloud (HCP)"]
        TFC["Terraform Cloud
API & Workspaces"]
        AgentPool["Agent Pool
(signalroom)"]
    end

    subgraph AWS["☁ AWS Cloud"]
        subgraph TFC_AGENT_VPC["TFC Agent VPC
var.vpc_cidr"]
            subgraph PUBLIC_SUBNETS["Public Subnets (Multi-AZ)"]
                IGW["Internet
Gateway"]
                NAT1["NAT Gateway
AZ-1"]
                NAT2["NAT Gateway
AZ-2"]
            end

            subgraph PRIVATE_SUBNETS["Private Subnets (Multi-AZ)"]
                subgraph ECS["ECS Fargate Cluster"]
                    TFCAgent1["TFC Agent
Container"]
                    TFCAgent2["TFC Agent
Container"]
                end

                subgraph AWS_ENDPOINTS["AWS VPC Endpoints"]
                    VPCE_SM["Secrets Manager
Endpoint"]
                    VPCE_CW["CloudWatch Logs
Endpoint"]
                    VPCE_ECR["ECR API/DKR
Endpoints"]
                    VPCE_S3["S3 Gateway
Endpoint"]
                end
```

```
                CONFLUENT_SG["Confluent PrivateLink
Security Group"]
            end

            DHCP["DHCP Options
(Custom DNS)"]
            TFC_AGENT_SG["TFC Agent
Security Group"]
        end

        subgraph TGW["Transit Gateway
signalroom-tgw"]
            TGWCore["TGW Core"]
            TGWRT["Route Table"]
        end

        subgraph DNS_VPC["DNS VPC (Centralized)
var.dns_vpc_cidr"]
            R53Inbound["Route53 Inbound
Resolver"]
            PHZ["Private Hosted Zones
*.aws.confluent.cloud"]
        end

        subgraph CLIENT_VPN_VPC["Client VPN VPC
var.client_vpn_vpc_cidr"]
            VPNEndpoint["Client VPN
Endpoint"]
        end

        subgraph WORKLOAD_VPCs["Workload VPCs
(Confluent PrivateLink)"]
            subgraph WL1["Workload VPC 1"]
                VPCE1["PrivateLink
Endpoint"]
            end
            subgraph WL2["Workload VPC N"]
                VPCEN["PrivateLink
Endpoint"]
            end
        end

        SecretsManager["AWS Secrets Manager
(TFC Agent Token)"]
        CloudWatch["CloudWatch Logs"]
        ECR_Registry["ECR Registry
(hashicorp/tfc-agent)"]
    end

    subgraph CONFLUENT["☁ Confluent Cloud"]
        PrivateLinkSvc["PrivateLink
Service"]
        Kafka["Kafka Cluster
(Private)"]
```

```
    end

    %% External Connections
    TFC <-->|"HTTPS/443
via NAT"| TFCAgent1
    TFC <-->|"HTTPS/443
via NAT"| TFCAgent2
    AgentPool -.->|"Agent Registration"| TFCAgent1

    %% Internal VPC Connections
    TFCAgent1 --> TFC_AGENT_SG
    TFCAgent2 --> TFC_AGENT_SG
    TFCAgent1 --> VPCE_SM
    TFCAgent2 --> VPCE_CW

    VPCE_SM -.->|"Private DNS"| SecretsManager
    VPCE_CW -.->|"Private DNS"| CloudWatch
    VPCE_ECR -.->|"Private DNS"| ECR_Registry

    NAT1 --> IGW
    NAT2 --> IGW
    TFCAgent1 -->|"0.0.0.0/0"| NAT1
    TFCAgent2 -->|"0.0.0.0/0"| NAT2

    %% DHCP & DNS Flow
    DHCP -->|"DNS Servers:
VPC + Centralized"| TFCAgent1
    TFCAgent1 -->|"DNS Query:
*.confluent.cloud"| R53Inbound

    %% Transit Gateway Connections
    TFC_AGENT_VPC -->|"TGW Attachment"| TGW
    DNS_VPC -->|"TGW Attachment"| TGW
    CLIENT_VPN_VPC -->|"TGW Attachment"| TGW
    WL1 -->|"TGW Attachment"| TGW
    WL2 -->|"TGW Attachment"| TGW

    %% Route Propagation
    TGWCore --> TGWRT

    %% DNS Resolution
    R53Inbound --> PHZ
    PHZ -->|"Returns Private IPs"| VPCE1

    %% PrivateLink Connections
    VPCE1 -->|"AWS PrivateLink"| PrivateLinkSvc
    VPCEN -->|"AWS PrivateLink"| PrivateLinkSvc
    PrivateLinkSvc --> Kafka

    %% TFC Agent to Workload VPCs
    TFC_AGENT_SG -->|"HTTPS/443
Kafka/9092"| CONFLUENT_SG
    CONFLUENT_SG -->|"via TGW"| VPCE1
    CONFLUENT_SG -->|"via TGW"| VPCEN
```

```
    %% Styling
    classDef tfcStyle fill:#5c4ee5,stroke:#3d32a8,stroke-
width:2px,color:#fff
    classDef vpcStyle fill:#e8f0fe,stroke:#1a73e8,stroke-width:2px
    classDef tgwStyle fill:#fef7e0,stroke:#f9ab00,stroke-width:3px
    classDef dnsStyle fill:#e6f4ea,stroke:#34a853,stroke-width:2px
    classDef confluentStyle fill:#f3e8fd,stroke:#9334e6,stroke-width:2px
    classDef endpointStyle fill:#fce8e6,stroke:#ea4335,stroke-width:1px
    classDef ecsStyle fill:#fff3e0,stroke:#ff9800,stroke-width:2px

    class TERRAFORM_CLOUD tfcStyle
    class TFC_AGENT_VPC,CLIENT_VPN_VPC,WORKLOAD_VPCs,WL1,WL2 vpcStyle
    class TGW tgwStyle
    class DNS_VPC dnsStyle
    class CONFLUENT confluentStyle
    class AWS_ENDPOINTS,VPCE_SM,VPCE_CW,VPCE_ECR,VPCE_S3 endpointStyle
    class ECS ecsStyle
```

### 4.2.1 Key Features Required for Confluent PNI to Work (TFC Agent Configuration)

#### 4.2.1.1 Custom DHCP Options for DNS Resolution

- DHCP Options Set configured with **dual DNS servers**: VPC default DNS (`cidrhost(vpc_cidr, 2)`) AND centralized DNS VPC resolver IPs
- Region-aware domain name configuration (`ec2.internal` for us-east-1, `{region}.compute.internal` for others)
- Associates TFC Agent VPC with custom DHCP options to route Confluent domain queries to the central DNS infrastructure

#### 4.2.1.2 Transit Gateway Connectivity

- TFC Agent VPC attached to shared Transit Gateway with DNS support enabled
- Explicit route table association and route propagation (not using TGW defaults)
- Routes added from private subnets to: DNS VPC, and Client VPN VPC
- Flattened route map pattern (`for_each`) ensures routes are created for every workload VPC CIDR

#### 4.2.1.3 Security Group Configuration for Kafka Traffic

- **TFC Agent Security Group** with egress rules for:
  - HTTPS (443) and Kafka (9092) to each workload VPC CIDR
  - DNS (UDP/TCP 53) to DNS VPC CIDR specifically
  - General HTTPS/HTTP for Terraform Cloud API and package downloads

#### 4.2.1.4 AWS VPC Endpoints for Private Service Access

- **Interface endpoints** with private DNS enabled for: Secrets Manager, CloudWatch Logs, ECR API, ECR DKR
- **S3 Gateway endpoint** (required for ECR image layer pulls)

- Dedicated security group for VPC endpoints allowing HTTPS from within VPC
- Eliminates NAT Gateway dependency for AWS service calls

#### 4.2.1.5 ECS Fargate Deployment Pattern

- TFC Agents run in private subnets with `assign_public_ip = false`
- NAT Gateways per AZ for outbound internet (Terraform Cloud API communication)
- Agent token stored in Secrets Manager, fetched via VPC Endpoint
- Container health checks and deployment circuit breaker for reliability

#### 4.2.1.6 IAM Permissions for Infrastructure Management

- Task role with Transit Gateway, VPC, Route53 Resolver, and Client VPN management permissions
- Execution role with Secrets Manager access for agent token retrieval
- KMS permissions scoped to Secrets Manager service for encryption/decryption

#### 4.2.1.7 Network Architecture Summary

- **Hub-and-spoke model**: TGW connects TFC Agent VPC → DNS VPC → Workload VPCs

---

## 5.0 Configuration

All sensitive values are passed as environment variables (never stored in `.tfvars`). The `deploy.sh` script handles setting `TF_VAR_*` exports automatically after AWS SSO authentication.

### 5.1 Key Input Variables

| Variable | Description |
|---|---|
| `tgw_id` | Existing Transit Gateway ID |
| `tgw_rt_id` | Transit Gateway Route Table ID |
| `vpn_vpc_id` | VPN VPC ID |
| `vpn_vpc_rt_ids` | Comma-separated VPN VPC route table IDs |
| `vpn_endpoint_id` | AWS Client VPN Endpoint ID |
| `vpn_target_subnet_ids` | Comma-separated VPN associated subnet IDs |
| `tfc_agent_vpc_id` | Terraform Cloud Agent VPC ID |
| `tfc_agent_vpc_rt_ids` | Comma-separated TFC Agent VPC route table IDs |
| `eni_number_per_subnet` | Number of ENIs per subnet (default: `17`) |
| `subnet_count` | Number of subnets to create per AZ (default: `3`) |
| `aws_region` | AWS region for all resources |

### 5.2 CIDR Allocations

| Network | CIDR |
|---|---|
| PNI Hub VPC | `10.3.0.0/20` |
| Sandbox Spoke VPC | `10.0.0.0/20` |
| Shared Spoke VPC | `10.1.0.0/20` |

All VPCs use 3 subnets across 3 AZs with `/4` new bits of sub-netting.

---

# 6.0 Deployment

## 6.1 Create

```
./deploy.sh create --profile=<SSO_PROFILE_NAME> \
                   --confluent-api-key=<CONFLUENT_API_KEY> \
                   --confluent-api-secret=<CONFLUENT_API_SECRET> \
                   --tgw-id=<TGW_ID> \
                   --tgw-rt-id=<TGW_RT_ID> \
                   --tfc-agent-vpc-id=<TFC_AGENT_VPC_ID> \
                   --tfc-agent-vpc-rt-ids=<TFC_AGENT_VPC_RT_IDs> \
                   --vpn-vpc-id=<VPN_VPC_ID> \
                   --vpn-vpc-rt-ids=<VPN_VPC_RT_IDs> \
                   --vpn-endpoint-id=<VPN_ENDPOINT_ID> \
                   --vpn-target-subnet-ids=<VPN_TARGET_SUBNET_IDs> \
                   --pni-hub-vpc-cidr=<PNI_HUB_VPC_CIDR>
```

Here's the argument table for `deploy.sh create` command:

| Argument | Required | Placeholder | Description |
|---|---|---|---|
| `--profile` | ✅ | `<SSO_PROFILE_NAME>` | AWS SSO profile name used to authenticate via `aws sso login` and derive the AWS region and account ID for the session. |
| `--confluent-api-key` | ✅ | `<CONFLUENT_API_KEY>` | Confluent Cloud API key used by Terraform to authenticate against the Confluent provider. |
| `--confluent-api-secret` | ✅ | `<CONFLUENT_API_SECRET>` | Confluent Cloud API secret paired with the API key for Terraform provider authentication. |

| Argument | Required | Placeholder | Description |
|---|---|---|---|
| `--tgw-id` | ✅ | `<TGW_ID>` | AWS Transit Gateway ID (e.g., `tgw-xxxxxxxxxxxxxxxxx`) that serves as the central hub routing traffic between the TFC Agent VPC, VPN VPC, and the PrivateLink Network Interconnect (PNI) Hub VPC. |
| `--tgw-rt-id` | ✅ | `<TGW_RT_ID>` | AWS Transit Gateway Route Table ID (e.g., `tgw-rtb-xxxxxxxxxxxxxxxxx`) to which routes are added for inter-VPC traffic through the Transit Gateway. |
| `--tfc-agent-vpc-id` | ✅ | `<TFC_AGENT_VPC_ID>` | VPC ID of the VPC hosting the Terraform Cloud Agent (ECS Fargate). This VPC is attached to the Transit Gateway for connectivity to the Confluent Private Network. |
| `--tfc-agent-vpc-rt-ids` | ✅ | `<TFC_AGENT_VPC_RT_IDs>` | Comma-separated list of route table IDs within the TFC Agent VPC. Routes to the PNI Hub VPC CIDR are injected into these tables to enable PrivateLink reachability. |
| `--vpn-vpc-id` | ✅ | `<VPN_VPC_ID>` | VPC ID of the VPC where the Client VPN endpoint is deployed, allowing operator/admin connectivity into the private network. |
| `--vpn-vpc-rt-ids` | ✅ | `<VPN_VPC_RT_IDs>` | Comma-separated list of route table IDs within the VPN VPC. Routes to the PNI Hub VPC CIDR are added here so VPN-connected clients can reach Confluent Private Network. |
| `--vpn-endpoint-id` | ✅ | `<VPN_ENDPOINT_ID>` | AWS Client VPN Endpoint ID (e.g., `cvpn-endpoint-xxxxxxxxxxxxxxxxx`). Used to associate target subnets and configure VPN routing rules. |
| `--vpn-target-subnet-ids` | ✅ | `<VPN_TARGET_SUBNET_IDs>` | Comma-separated list of subnet IDs within the VPN VPC to which the Client VPN endpoint is associated, determining which AZs VPN traffic can ingress through. |

| Argument | Required | Placeholder | Description |
|---|---|---|---|
| `--pni-hub-vpc-cidr` | ✅ | `<PNI_HUB_VPC_CIDR>` | CIDR block of the Confluent PrivateLink Network Interconnect (PNI) Hub VPC (e.g., `10.x.x.x/xx`). This CIDR is injected as a route destination into the TFC Agent and VPN VPC route tables so traffic destined for Confluent endpoints is properly directed through the Transit Gateway. |

> All `12` arguments are required, the script exits with code 85 if any are missing.

The script will:

1. Authenticate to AWS SSO and export temporary credentials.
2. Export all `TF_VAR_*` environment variables.
3. Run `terraform init`, `terraform plan`, prompt for confirmation, then `terraform apply`.
4. Generate a Terraform graph visualization at `docs/images/terraform-visualization.png`.

### 6.2 Destroy

```
./deploy.sh destroy --profile=<SSO_PROFILE_NAME> \
                    --confluent-api-key=<CONFLUENT_API_KEY> \
                    --confluent-api-secret=<CONFLUENT_API_SECRET> \
                    --tgw-id=<TGW_ID> \
                    --tgw-rt-id=<TGW_RT_ID> \
                    --tfc-agent-vpc-id=<TFC_AGENT_VPC_ID> \
                    --tfc-agent-vpc-rt-ids=<TFC_AGENT_VPC_RT_IDs> \
                    --vpn-vpc-id=<VPN_VPC_ID> \
                    --vpn-vpc-rt-ids=<VPN_VPC_RT_IDs> \
                    --vpn-endpoint-id=<VPN_ENDPOINT_ID> \
                    --vpn-target-subnet-ids=<VPN_TARGET_SUBNET_IDs> \
                    --pni-hub-vpc-cidr=<PNI_HUB_VPC_CIDR>
```

Here's the argument table for `deploy.sh destroy` command:

| Argument | Required | Placeholder | Description |
|---|---|---|---|
| `--profile` | ✅ | `<SSO_PROFILE_NAME>` | AWS SSO profile name used to authenticate via `aws sso login` and derive the AWS region and account ID for the session. |
| `--confluent-api-key` | ✅ | `<CONFLUENT_API_KEY>` | Confluent Cloud API key used by Terraform to authenticate against the Confluent provider. |

| Argument | Required | Placeholder | Description |
|---|---|---|---|
| `--confluent-api-secret` | ✅ | `<CONFLUENT_API_SECRET>` | Confluent Cloud API secret paired with the API key for Terraform provider authentication. |
| `--tgw-id` | ✅ | `<TGW_ID>` | AWS Transit Gateway ID (e.g., `tgw-xxxxxxxxxxxxxxxxx`) that serves as the central hub routing traffic between the TFC Agent VPC, VPN VPC, and the PrivateLink Network Interconnect (PNI) Hub VPC. |
| `--tgw-rt-id` | ✅ | `<TGW_RT_ID>` | AWS Transit Gateway Route Table ID (e.g., `tgw-rtb-xxxxxxxxxxxxxxxxx`) to which routes are added for inter-VPC traffic through the Transit Gateway. |
| `--tfc-agent-vpc-id` | ✅ | `<TFC_AGENT_VPC_ID>` | VPC ID of the VPC hosting the Terraform Cloud Agent (ECS Fargate). This VPC is attached to the Transit Gateway for connectivity to the Confluent Private Network. |
| `--tfc-agent-vpc-rt-ids` | ✅ | `<TFC_AGENT_VPC_RT_IDs>` | Comma-separated list of route table IDs within the TFC Agent VPC. Routes to the PNI Hub VPC CIDR are injected into these tables to enable PrivateLink reachability. |
| `--vpn-vpc-id` | ✅ | `<VPN_VPC_ID>` | VPC ID of the VPC where the Client VPN endpoint is deployed, allowing operator/admin connectivity into the private network. |
| `--vpn-vpc-rt-ids` | ✅ | `<VPN_VPC_RT_IDs>` | Comma-separated list of route table IDs within the VPN VPC. Routes to the PNI Hub VPC CIDR are added here so VPN-connected clients can reach Confluent Private Network. |
| `--vpn-endpoint-id` | ✅ | `<VPN_ENDPOINT_ID>` | AWS Client VPN Endpoint ID (e.g., `cvpn-endpoint-xxxxxxxxxxxxxxxxx`). Used to associate target subnets and configure VPN routing rules. |

| Argument | Required | Placeholder | Description |
|---|---|---|---|
| `--vpn-target-subnet-ids` | ✅ | `<VPN_TARGET_SUBNET_IDs>` | Comma-separated list of subnet IDs within the VPN VPC to which the Client VPN endpoint is associated, determining which AZs VPN traffic can ingress through. |
| `--pni-hub-vpc-cidr` | ✅ | `<PNI_HUB_VPC_CIDR>` | CIDR block of the Confluent PrivateLink Network Interconnect (PNI) Hub VPC (e.g., `10.x.x.x/xx`). This CIDR is injected as a route destination into the TFC Agent and VPN VPC route tables so traffic destined for Confluent endpoints is properly directed through the Transit Gateway. |

> All `12` arguments are required, the script exits with code 85 if any are missing.

Destroy runs `terraform destroy -auto-approve` and regenerates the visualization.

---

## 7.0 Outputs

| Output | Description |
|---|---|
| `confluent_pni_hub_gateway_id` | ID of the `confluent_gateway` resource (PNI Hub) |
| `confluent_pni_hub_access_point_id` | ID of the `confluent_access_point` resource |
| `confluent_environment_id` | ID of the Confluent Cloud environment where clusters are provisioned |
| `confluent_sandbox_cluster_id` | ID of the sandbox Kafka cluster (confluent_kafka_cluster) |
| `confluent_shared_cluster_id` | ID of the shared Kafka cluster (confluent_kafka_cluster) |

---

## 8.0 Security Design

**Security Group (PNI ENIs):** Ingress-only on ports `443` (HTTPS/REST/Schema Registry) and `9092` (Kafka), sourced from the PNI Hub VPC CIDR, TFC Agent VPC CIDR, VPN VPC CIDR, and Client VPN CIDR. **No egress rules are defined**, which causes Terraform to revoke AWS's default `0.0.0.0/0` egress — intentionally mirroring PrivateLink's unidirectional behavior and preventing Confluent-initiated connections into the customer network.

**Network ACL:** Allows TCP ingress on `443`, `9092`, and ephemeral ports `1024-65535`. Allows all egress.

**ENI Permissions:** `aws_network_interface_permission` grants Confluent's AWS account `INSTANCE-ATTACH` permission on each customer-owned ENI. This is the core PNI handshake — Confluent attaches its broker VMs to your ENIs without your traffic ever leaving the AWS backbone.

## 9.0 How PNI Differs from PrivateLink

| Aspect | PrivateLink | PNI |
|---|---|---|
| ENI ownership | Confluent's account | **Customer's account** |
| DNS | Requires PHZ + VPC associations | Confluent manages DNS |
| Connectivity model | VPC Interface Endpoint | ENIs registered via `confluent_access_point` |
| Egress control | Unidirectional by design | Explicit empty egress on SG required |
| Port 53 (DNS) rules | Required in SG | **Not needed** |

## 10.0 Resources

- Confluent PNI Documentation
- Confluent PNI FAQ
- AWS Multi-VPC ENI Attachment
- confluent_gateway Terraform resource
- confluent_access_point Terraform resource